

PHYCON-26 Poster Presentation Schedule

Day 1 – 28th May 2026

(Condensed Matter Physics & Material Science)

Sr. No.	Name of the Presenter	Poster Title	Poster ID	Affiliation
1	Simranjit Kaur	Tuning the Properties of Titanium dioxide Nanoparticles through Aluminum Doping	CMP-PP-01	NIT Hamirpur
2	Astha Thakur	Spectroscopic Ellipsometric Study of Al-doped ZnO (AZO) grow on silicon substrate	CMP-PP-02	NIT Hamirpur
3	Medha Devgan	Realization of Broadband Terahertz Metamaterial Absorber Using Photosensitive Gallium Arsenide	CMP-PP-03	Guru Nanak Dev University, Amritsar
4	Rajeshwar Raj Singh	Design and Optimization of Ti O 2 – ZnO–Ti O 2 Multilayer Anti Reflection Coating Using Transfer Matrix Method	CMP-PP-04	NIT Hamirpur
5	Agathyan B.	Phonon-Mediated Optical Response in Monolayer SnS 2 : A First-Principles Many-Body Perturbation Theory Study	CMP-PP-05	Pondicherry University
6	Jiya	Universal Ohmic Contacts to β -Ga ₂ O ₃ using Interfacial Dipoles of ultra-thin MgF ₂ layer and related defect passivation	CMP-PP-06	IIT Ropar
7	Bhavay Gakhar	Compositionally Tuned CoFe-LDH Electrocatalyst for Energy Efficient Alkaline Water Electrolysis	CMP-PP-07	IIT Ropar
8	Satyam Shankhdhar	Harnessing the mechanical and magnetic energy with PMN-PT/Ni-Mn-In-based flexible piezoelectric nanogenerator	CMP-PP-08	IIT Roorkee
9	Vijay	Trace-level simultaneous monitoring of multiple heavy metal ions using	CMP-PP-09	IIT Roorkee

		AlScN-based flexible electrochemical sensor		
10	Tanya Chaudhary	Revealing the Potential of 2D WS ₂ Memristors as an Artificial Synapse with Resilient Gradual Behavior at High Temperatures for Neuromorphic Applications	CMP-PP-10	IIT Roorkee
11	Sonali Dehariya	Kelvin-Helmholtz Instability in a two component dipolar Bose-Einstein condensate with tilted polarization	CMP-PP-11	IIT Ropar
12	Nidhi	Energy-Efficient AlScN-based Flexible Artificial Synapse for High-Density Storage and Neuromorphic Computing Applications	CMP-PP-12	IIT Roorkee
13	Urvesh Prajapati	Unveiling the Multifunctionality of Flexible Nb ₂ O ₅ -based Memristor Emulating Synaptic Behavior for Neuromorphic Computing	CMP-PP-13	IIT Roorkee
14	Vikas	A First-Principles Multi-Property Optimization of Halogen-Substituted PEA ₂ PbI ₄ Layered hybrid Perovskites	CMP-PP-14	IIT Ropar
15	Manpreet Kaur	Energy-Efficient WS ₂ /Si Heterojunction Photodetector with Enhanced UV-NIR Broadband Behavior for Advanced Optoelectronic Devices.	CMP-PP-15	IIT Roorkee
16	Tanisha Burman	Leveraging Higher Order Mode for Magnetic Field Sensing in AlN-based Single-Port Delay Line Resonator	CMP-PP-16	IIT Roorkee
17	Gagandeep Singh	Highly Efficient, Field-Free Spin–Orbit Torque Switching in 2D Fe _{5–x} Co _x GeTe ₂ /Weyl Semimetal Heterostructures	CMP-PP-17	IIT Ropar
18	Priya Rani	PtSe ₂ /MoSe ₂ Heterostructure for Self-Powered Ultrafast Polarization-Sensitive Photodetection	CMP-PP-18	IIT Ropar
19	Rohit Dahiya	Self-powered solar-blind photodetector enabled by controlled reducing annealing of α-Ga ₂ O ₃	CMP-PP-19	IIT Ropar

20	Himanshi Sharma	Observation of Negative Magnetoimpedance and Positive Magnetodielectric Properties in Ferromagnetic $\text{La}_{1.4}\text{Sr}_{1.6}\text{Mn}_2\text{O}_7$ – $\text{La}_{0.67}\text{Sr}_{0.33}\text{MnO}_3$ Composite	CMP-PP-20	Central University of Rajasthan, Ajmer
21	Komal Bhadrapati	Effect of Calcination Temperature on the Magnetic and Transport Properties of Nanocrystalline $\text{La}_{0.7}\text{Sr}_{1.3}\text{MnO}_4$ Manganite	CMP-PP-21	Central University of Rajasthan, Ajmer
22	Gajendra Kumar Saini	Structural and Magnetic Properties of $\text{Al}_2\text{Mn}_{1-x}\text{Fe}_{1+x}$ ($x = 0.4, 0.2$) Saini Anti-Heusler Alloys	CMP-PP-22	Central University of Rajasthan, Ajmer

(Light Matter Interaction & Quantum Information)

Sr. No.	Name of the Presenter	Poster Title	Poster ID	Affiliation
1	Priyanshi Gupta	Characterization of the Phase Sensitivity in Linear and Nonlinear Interferometers	LMI-PP-01	IIT Ropar
2	Tanuja Tripathi	Design and Analysis of Multilayer Optical Filter	LMI-PP-02	NIT Hamirpur
3	Priyanka Bhasker	Mechanochemical assisted thermal deformation of Bi_2O_3 / Bi_2WO_6 sheets to octahedral structures for effective photoelectrochemical performance	LMI-PP-03	Central University of Rajasthan
4	Shivani Semwal	Thermally invariant Dielectric and plasmonic devices for Integrated Nanophotonics	LMI-PP-04	IIT Ropar
5	Karthik Kumar Pradeep	Longitudinal spin relaxation in ensembles of nitrogen-vacancy centers in diamond	LMI-PP-14	IIT Ropar
6	Priya	Experimental Determination of Average M-shell Fluorescence Yield for ^{90}Th	LMI-PP-07	Punjabi University, Patiala
7	Pooja Chugh	Generation of sub-micron thick liquid sheet using single jet impingement on knife-edge	LMI-PP-08	IISER Mohali
8	Sania Ayoub	Quantum thermometry using qubits as a probe	LMI-PP-09	IIT Ropar

9	Diksha Maurya	Hybrid Nanocavity coupling for enhanced NV center zero-phonon line emission	LMI-PP-18	IIT Ropar
10	Sunny	Generating Needle Beam of Single Photons with a Flat Lens	LMI-PP-11	IIT Ropar
11	Amarnath	Nanoscale Optomechanical Droplet Modes	LMI-PP-12	IISER Mohali

(Nuclear and Particle Physics)

Sr. No.	Name of the Presenter	Poster Title	Poster Code	Affiliation
1	Dr. Sahab Singh	High-Spin Spectroscopy of ^{201}Po	NP-OP-01	IIT Ropar
2	Harshik	Multimodal fission of $^{201}\text{Tl}^*$ via cluster preformation	NP-OP-02	Central University of Himachal Pradesh
3	Dr. Amanjot	Fabrication of carbon-backed ^{191}Ir thin film targets for the study of low-energy nuclear reaction dynamics	NP-OP-03	IIT Ropar
4	Rabina Goswami	Characterization of p-type HPGe Detector for Terrestrial Radioactivity Measurements	NP-OP-04	IIT Ropar
5	Dr. Anjali Rani			

Day 2 – 29th May 2026

(Condensed Matter Physics & Material Science)

Sr. No.	Name of the Presenter	Poster Title	Poster ID	Affiliation
1	Chetan Kumawat	Structural, Magnetic and Transport Properties of Al_2MnTi Anti-Heuser Alloy	CMP-PP-23	NIT Hamirpur
2	Abhishek Kumar	Synergistic External Stimuli Induced Electronic Modulation in Interlayer Misregistered Black Arsenic-Black Phosphorous van der Waals Heterostructure	CMP-PP-24	IIT Ropar
3	Vinit	Room Temperature Wafer-Scale Growth Of 1T' MoTe_2 For Self-Powered Broadband Photodetector	CMP-PP-25	IIT Ropar

4	Shiksha	Two-dimensional materials and their heterostructure-based memristor	CMP-PP-26	IIT Ropar
5	Moumita Pradhan	Cu-Doped Ga ₂ O ₃ Thin Films Enabling High-Performance p-Type Solar-Blind Photodetector	CMP-PP-27	IIT Ropar
6	Suman	Electronic properties of Co substituted FePO ₄ from extended Hubbard functionals	CMP-PP-28	IIT Ropar
7	Jatin Sagar	Synthesis and Investigation of Structural and Optical Characteristics of ZnO and Al-Doped ZnO Nanopowders	CMP-PP-29	NIT Hamirpur
8	Dharmender	Unveiling In ₂ ZnS ₅ Monolayer as a Promising Cathode for Next-Generation Alkali-Metal Ion Batteries	CMP-PP-30	IIT Ropar
9	Abhimanyu Sharma	Investigation of coupling regime in cavity magnonics	CMP-PP-31	Central University of Haryana
10	Chirag Kalouni	Large Damping like torque in Permalloy/Cr* bilayer system	CMP-PP-32	IIT Ropar
11	Abhishek Kumar	Demonstration of field-free switching via Orbital Hall effect in NiFe/Cr bilayer	CMP-PP-33	IIT Ropar
12	Shikha	Designing high performance doped Nickel (001) electrocatalyst for Oxygen Evolution Reaction(OER) through water splitting.	CMP-PP-34	IIT Ropar
13	Prineeth Jeethan	Investigation of Self-Inverse Spin Hall Effect in Symmetric MgO/NiFe/MgO Heterostructures	CMP-PP-35	IIT Ropar
14	Ashriwad Patra	Biaxial Stress Effects on the Electronic Properties of the Equiatomic Quaternary Heusler Alloy TiCrMnSb: A First Principles Study	CMP-PP-36	NIT Hamirpur
15	Pawan Kumar	Fluence-Dependent Morphological Evolution of PDMS under Low-Energy Argon Ion Bombardment	CMP-PP-37	IIT Ropar
16	Neha	Fabrication and Characterization of Laser-Induced Graphene on	CMP-PP-38	IIT Ropar

		Polyimide Using Nd:YAG and CO ₂ Laser Scribing		
17	Ayub Sekh	Self-Learning Gaussian Process Framework for Parameter Optimization in APCVD Growth of WS ₂	CMP-PP-39	IIT Ropar
18	Alka Devi	Tailoring Voltage-Dependent Phase Transition Dynamics in Blue Phase Liquid Crystals through MWCNT Doping	CMP-PP-40	DAV University, Jalandhar
19	Archita	Synthesis, Structural & Spectral Studies of Eu ³⁺ -Doped KSrB ₅ O ₉ /PVP Nanocomposite Thin Film on Glass Substrate	CMP-PP-41	SMVDU, Katra
20	Meniak Khajuria	Synthesis, Structural and Spectral Studies of Er ³⁺ Doped LiSrVO ₄ Nanophosphor	CMP-PP-42	SMVDU, Katra
21	Rohit Raina	Photoluminescence and Band Structure Studies of Eu ³⁺ Doped KSrB ₅ O ₉ Phosphor	CMP-PP-43	SMVDU, Katra

(Light Matter Interaction & Quantum Information)

Sr. No.	Name of the Presenter	Poster Title	Poster ID	Affiliation
01	Sanjoy Dutta	Quantum steganographic protocols using degenerate and entanglement-assisted quantum codes	LMI-PP-13	Proornaprajna Institute of Scientific Research
02	Kondeti Shiva Narassimha	Inverse-Designed 2.5 × 2.5 μm ² CNOT Quantum Logic Gate using Crystalline Se ₂ Sb ₃	LMI-PP-15	CSIR-CSIO, Chandigarh
03	Ashish	Bright and Ultra-Pure Single-Photon Emission from NV Center in Diamond	LMI-PP-16	IIT Ropar
04	Varun Rathour	Generation of Partially Coherent Fields with Tunable Spatial Correlations Using Stimulated Parametric Down-Conversion	LMI-PP-17	IIT Ropar
05	Rajneesh Fulara	Ground-State Search via Intrinsic Feedback Annealing in Laser Arrays	LMI-PP-19	IIT Ropar

06	Dr. Anil Kumar	Evaluation of L to M Shell Vacancy Transfer Probabilities for Bismuth	LMI-PP-20	Punjabi University, Patiala
07	Bhavana Pareek	Dual transformation of BiOCl sheets into faceted particles of Bi ₂ Sn ₂ O ₇ for multifunctional application	LMI-PP-21	Central University of Rajasthan
08	Ramanpreet Kaur	Synchrotron based measurements of L _i (i= 1-3) subshell fluorescence yields (ω_i) for hafnium	LMI-PP-22	Punjabi University
09	Dr. Neethu K	Generation of Uniform-Intensity Beams with Tunable Coherence in a Degenerate Cavity Laser	LMI-PP-23	IIT Ropar
10	Aditya Narayana Jena	Extending Depth-of-Focus of Femtosecond Pulses for Ultrafast Laser Micromachining	LMI-PP-24	IIT Ropar
11	Love Kumar Sharma	From Condensation to Negative-Temperature States: Index Controlled Multimode Soliton Dynamics	LMI-PP-05	IIT Ropar
12	Sitara Khan Teli	Mechanistic Insight of Raoult's Law for Evaporation Process in a Binary Liquid Mixture	LMI-PP-06	IIT Ropar
13	Sagar Shayer Yaar	Exploring Lipid and Surfactant Interactions with Self-Assembled Amino Acids at the Air–Aqueous Interface via Nonlinear Sum Frequency Generation Spectroscopy	LMI-PP-10	IIT Ropar

(Nuclear and Particle Physics)

Sr. No	Name of the Presenter	Poster title	Poster ID	Affiliation
14	Kanchan	Determination of the L ₃ Subshell Fluorescence Yield of Gadolinium Using Synchrotron Radiation	NP-PP-05	Punjabi University
15	Dr. Anjali Rani	Examination of systematic uncertainties in the extraction of Fierz interference term within the bSTILED project	NP-PP-07	ENSICAEN, LPC CAEN, FRANCE